Fiscal Year:	FY 2015	**Task Last Updated:**	FY 05/18/2015
PI Name:	Riascos-Castaneda, Roy M.D.		
Project Title:	Association of Diffusion Tensor Imaging Parameters of Optic Tracts and Cerebral White Matter Tracts with Visual Impairment and Structural Changes of the Eyes and Optic Nerves in Long-Duration Microgravity Exposure		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	Roy.F.Riascos@uth.tmc.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	713-704-1704
Organization Name:	University of Texas Health Science Center, Houston		
PI Address 1:	Department of Diagnostic & Interventional Imaging, Division of Neuroradiology		
PI Address 2:	6431 Fannin, MSB 2.130B		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77030	**Congressional District:**	9
Comments:	NOTE: Formerly at University of Texas Medical Branch, Galveston, until January 2015.		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2011 Crew Health NNJ11ZSA002NA
Start Date:	04/01/2013	**End Date:**	02/13/2015
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Villarreal, Jennifer	**Contact Phone:**	281-483-7306
Contact Email:	jennifer.v311larreal@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 2/13/2015 due to PI move, per NSSC information and JSC HRP (Ed., 5/6/15) NOTE: Extended to 3/30/2015, from original end date of 3/30/2014, per J. Villarreal/JSC and NSSC information (Ed., 3/18/14)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Alperin, Noam (University of Miami, Medical Campus) Barr, Yael (University of Texas Medical Branch, Galveston) Hakimelahi, Reza (University of Texas Medical Branch, Galveston) Hasan, Khader (University of Texas Health Science Center at Houston) Kramer, Larry (University of Texas Health Science Center at Houston) Sargsyan, Ashot (Wyle Laboratories, Inc.)		
Grant/Contract No.:	NNX13AH23G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Some crewmembers on long-duration missions have experienced visual impairment and concomitant anatomical changes in the eye detected by ophthalmoscopy and diagnostic imaging. Some of these changes have been reversible and others have persisted for months or even years, and have included increased optic nerve sheath diameter, tortuosity of the optic nerve, and flattening of the posterior globe. The optic nerves are direct extension of the encephalon; hence some of the changes in the eyes and optic nerves may represent changes occurring in the brain. In the terrestrial environment, similar findings have been described in cases of intracranial hypertension. Subtle changes in the intracranial pressure (ICP) can remain undetected if only anatomical-based imaging is used. Advanced MRI technology, often termed functional MRI, allows seeing beyond the gross anatomy to characterize brain pathways and function; these include tools with the ability to detect and image the motion of fluid around the brain tracts, known as Diffusion Tensor Imaging (DTI). DTI is based on the non isotropic (anisotropic) diffusion of water molecules if exposed to large magnetic fields; water molecules diffuse favoring one direction over the others. Through computer algorithms, a "diffusion tensor" is calculated for each volume element (voxel), allowing to identify voxels representing a given nerve fiber tract. One of the advantages of this system is that it can also detect the magnitude of decrease in signal in the anisotropic direction and calculate the fraction of anisotropy (FA) for a tract. Recently, DTI has been used to evaluate white matter alteration in various brain pathologies, including those with elevated ICP. The corticospinal tract (CST) is the pathway most frequently evaluated by DTI in different conditions, and is especially useful in hydrocephalus. For instance, Hattori et al. have shown that patients with idiopathic normal-pressure hydrocephalus (iNPH) have altered microstructures in the CST detected by DTI. Optic nerve tracts have also been studied with DTI for a variety of pathologies such as glaucoma, neuritis, and multiple sclerosis (MS). In a recent study on 104 MS patients, DTI abnormalities (a drop in FA) were found to be present and correlated with both loss of visual acuity and retinal nerve fiber layer thinning. Thus, DTI can provide functional information of the brain tracts and help detect early changes of the brain before they manifest anatomically. We suggest that DTI abnormalities are likely to be present in individuals subjected to long-duration microgravity, especially those with prolonged exposure to elevated ICP. Using the proposed technology, the project aims to detect changes in the DTI of the brain in individuals that have been exposed to microgravity and to compare them to preflight imaging and with clinical data and other parameters obtained in the process of crewmember monitoring for ICP elevation. This proposal will initiate the first study to evaluate brain microstructures in astronauts beyond anatomical characterization and CSF dynamics, and may offer much improved understanding of the mechanism of ICP elevation and vision changes in a subset of space crewmembers. Quantitative DTI data may also reflect the long-term cumulative effects of multiple microgravity exposures, which is a critical component of knowledge needed to enable human space missions. A 3-Tesla MRI protocol will be used to scan the brain of the study subjects using isovolumetric voxels and 20 diffusion tensor directions. One preflight and one postflight data acquisition will be reviewed to consider the degree of persistence of the quantitative DTI changes in the affected areas of the brain and in optic nerves. These findings will be compared to the astronaut’s clinical data and to non-invasive MRI based measurement of intracranial compliance and pressure measurements calculated from the craniospinal CSF fluid dynamics. ED. NOTE (May 2015): PI moved to University of Texas Health Science Center, Houston; grant is continued with the same title, grant number NNX15AE25G.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	NOTE (May 2015): PI moved to University of Texas Health Science Center, Houston; grant is continued with the same title, grant number NNX15AE25G. See that task for subsequent reporting.
Bibliography Type:	Description: (Last Updated: 08/25/2020)